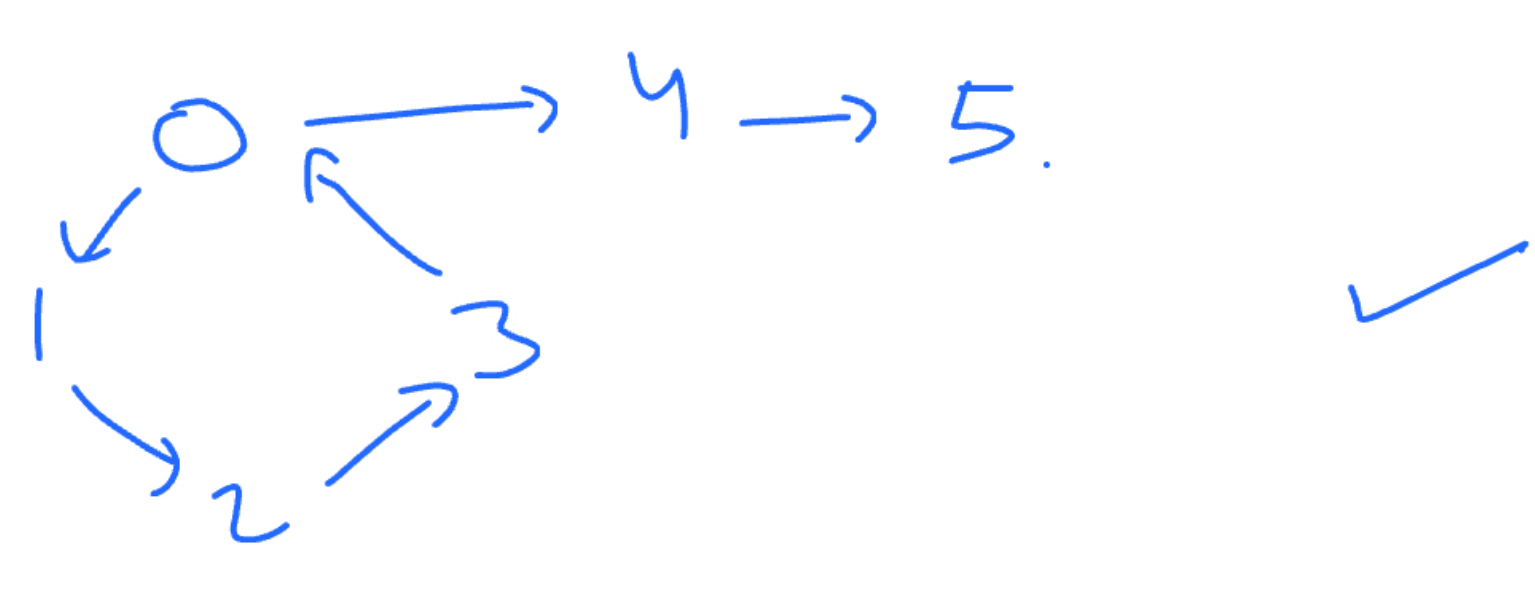
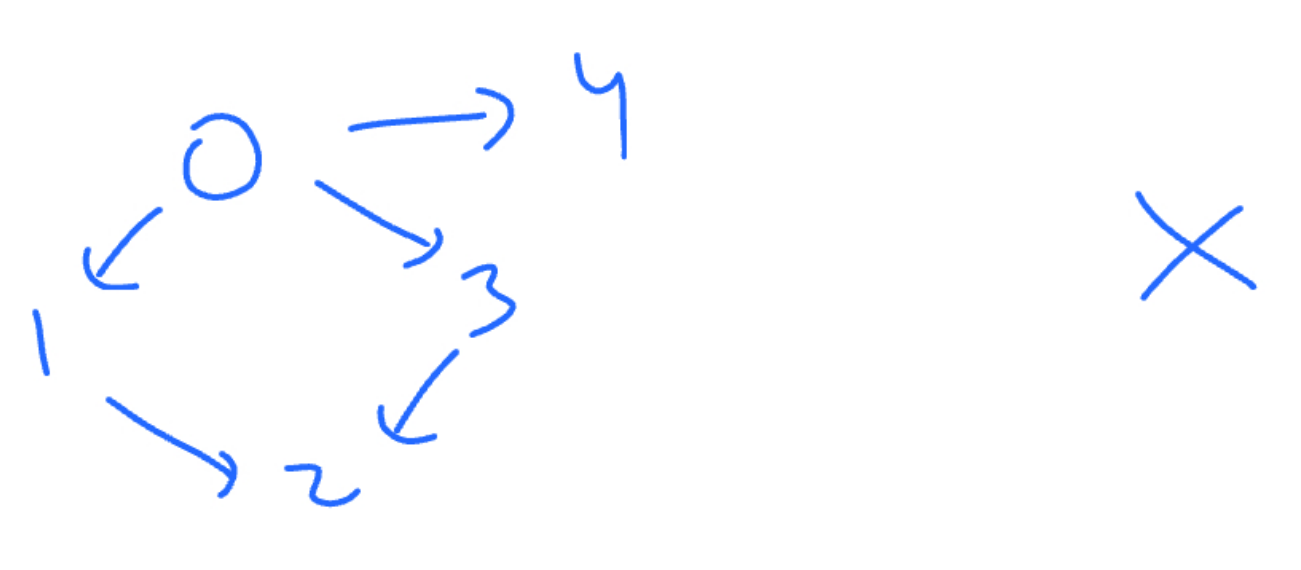
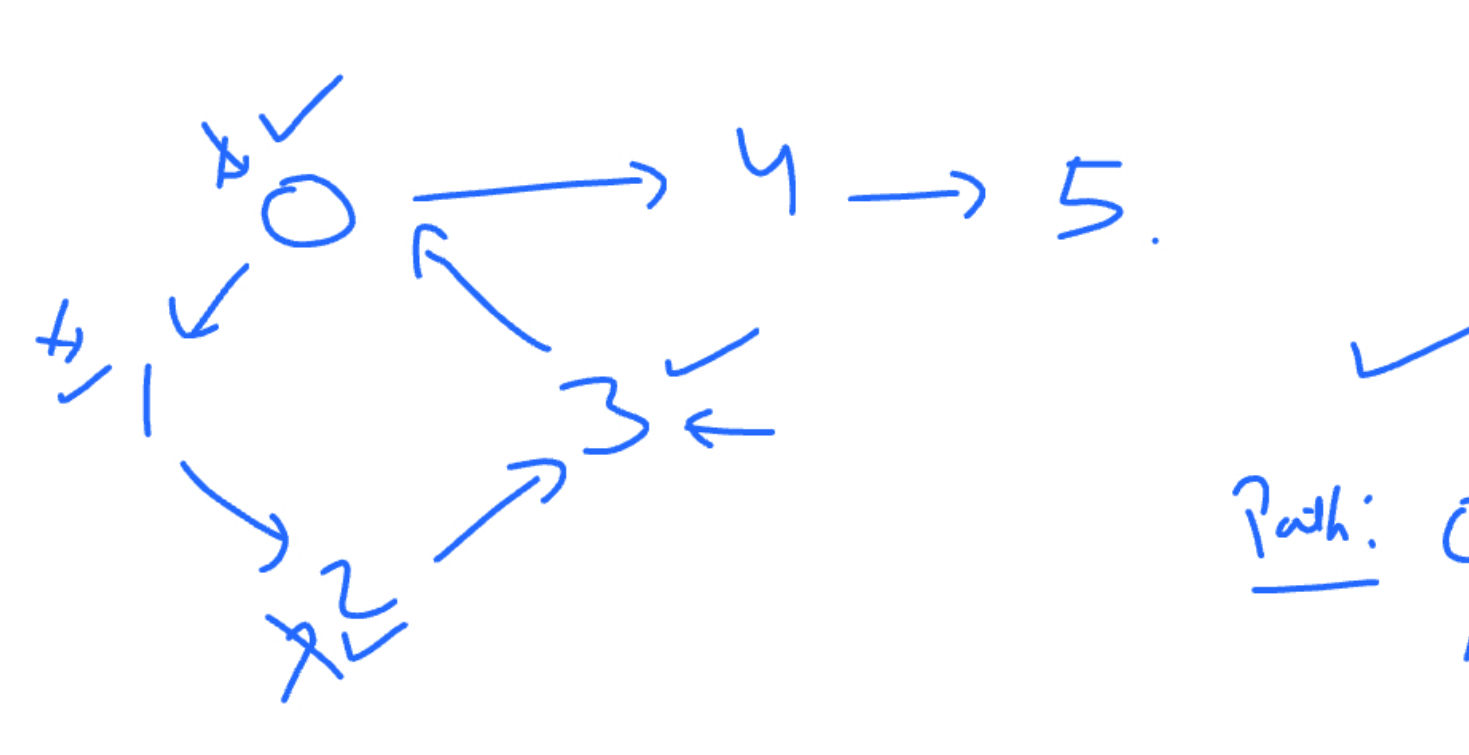
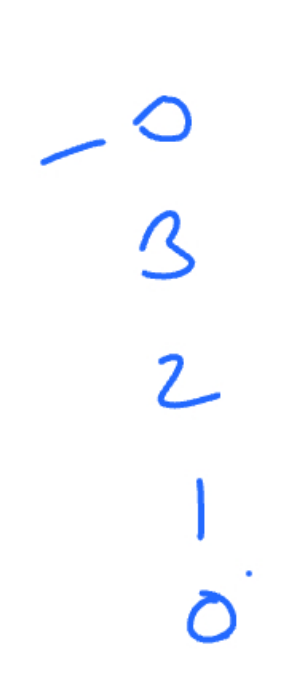
Directed Graph Cycle.

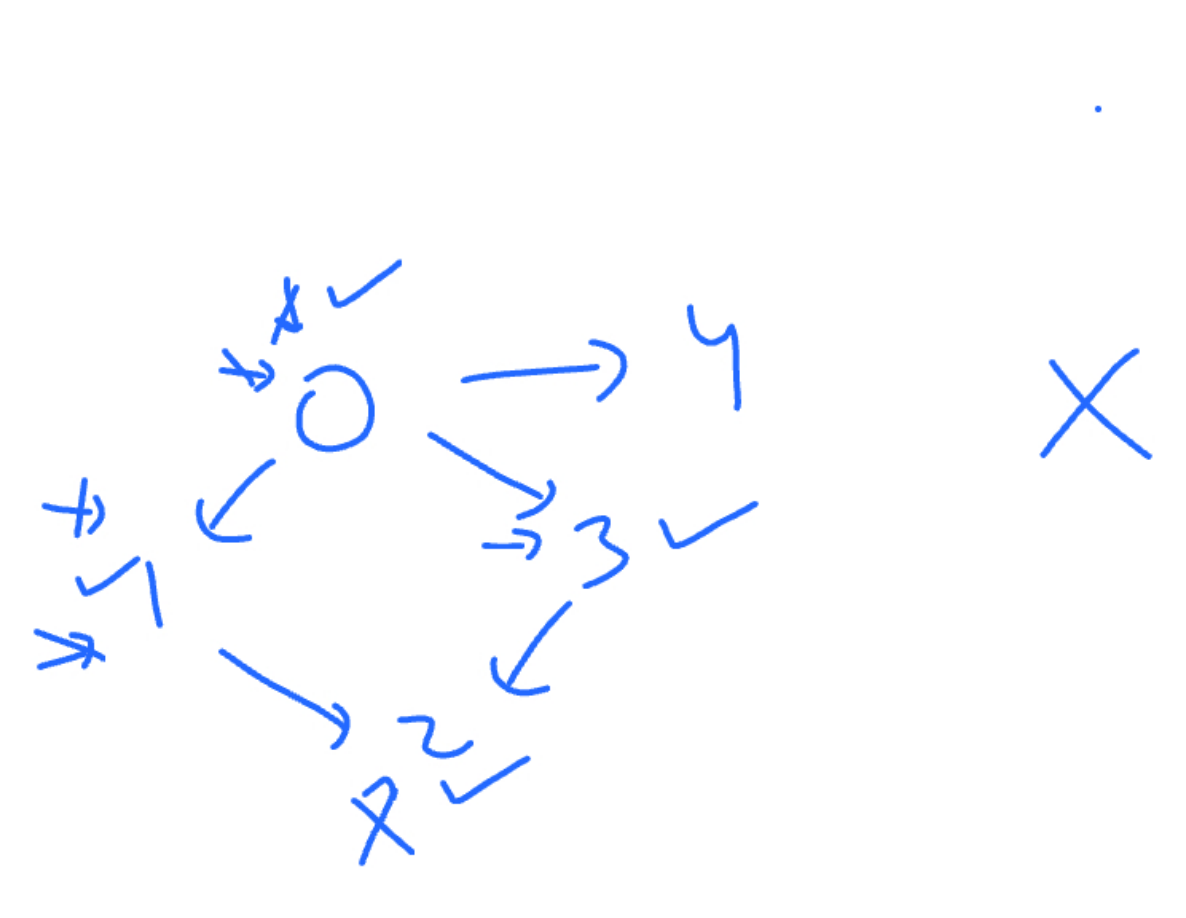






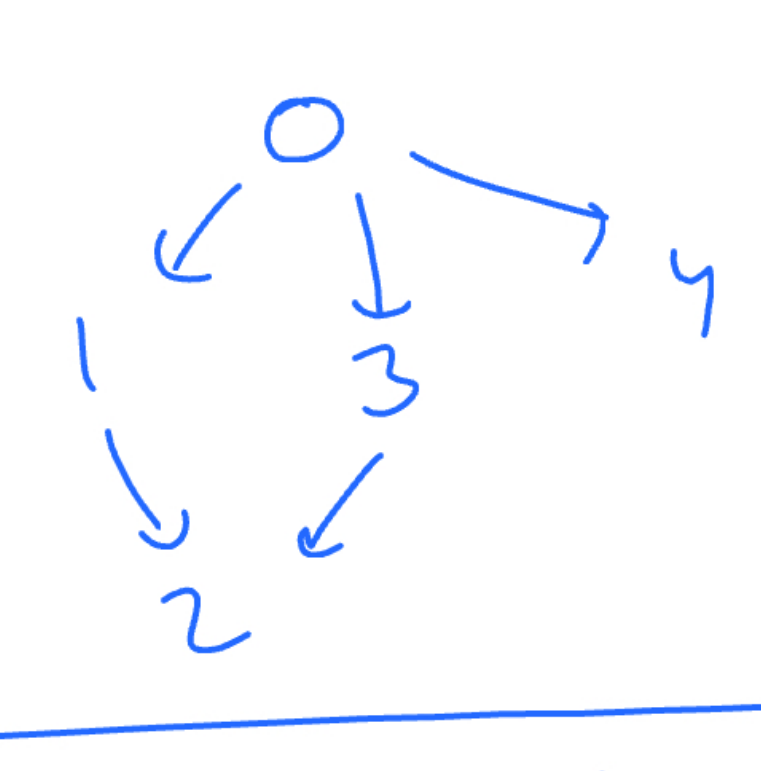
Path: 0→1→2→3





Path: 0→1→2

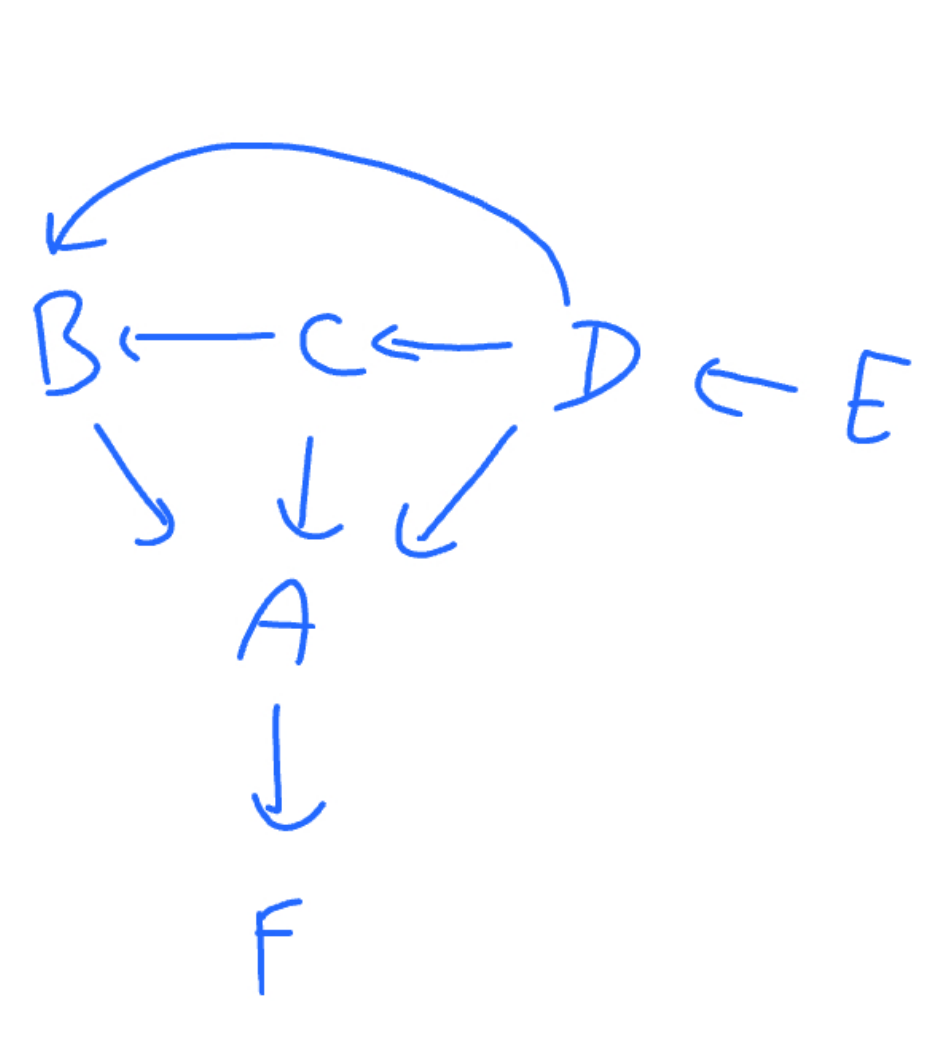
0→3→2



Topological sort.

U → V] → U should come before V

course.	Prereq.
→ A :	B, C, D
B :	C, D
C :	D
D :	E
E :	
F :	A

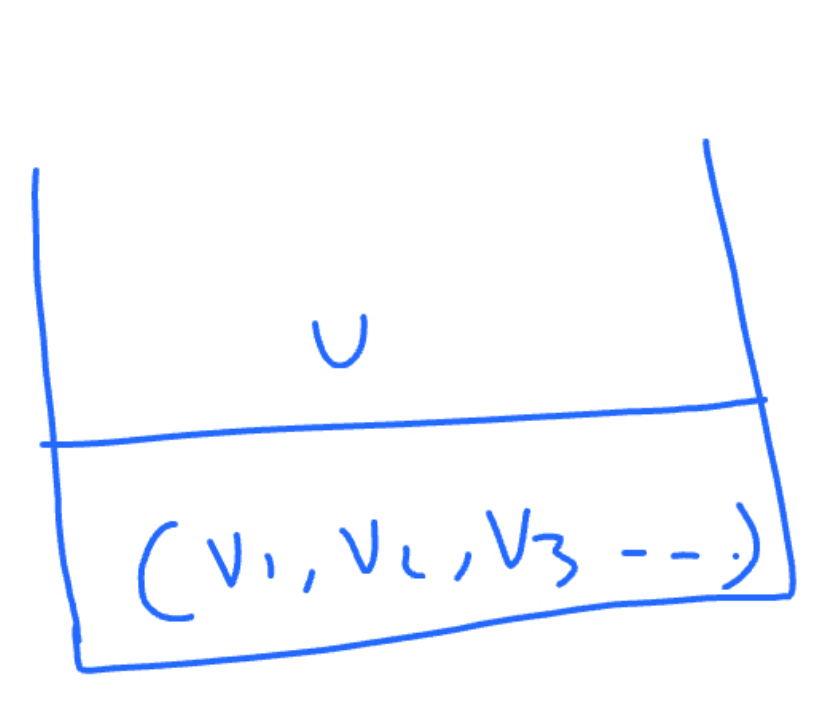


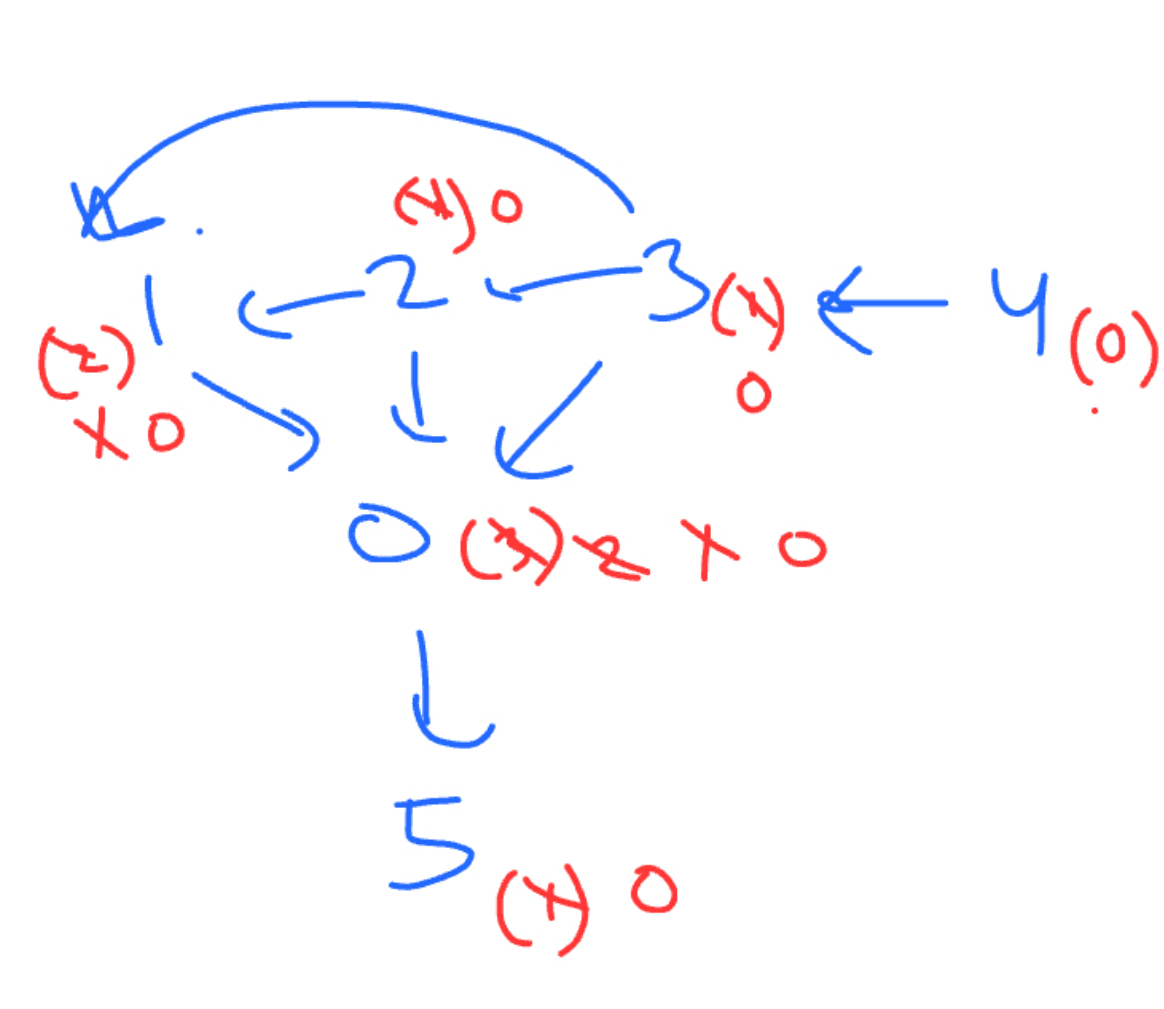
"E, D, C, B, A, F"

U

→ V₁, V₂, V₃ - - - .

→ stack. push(U)

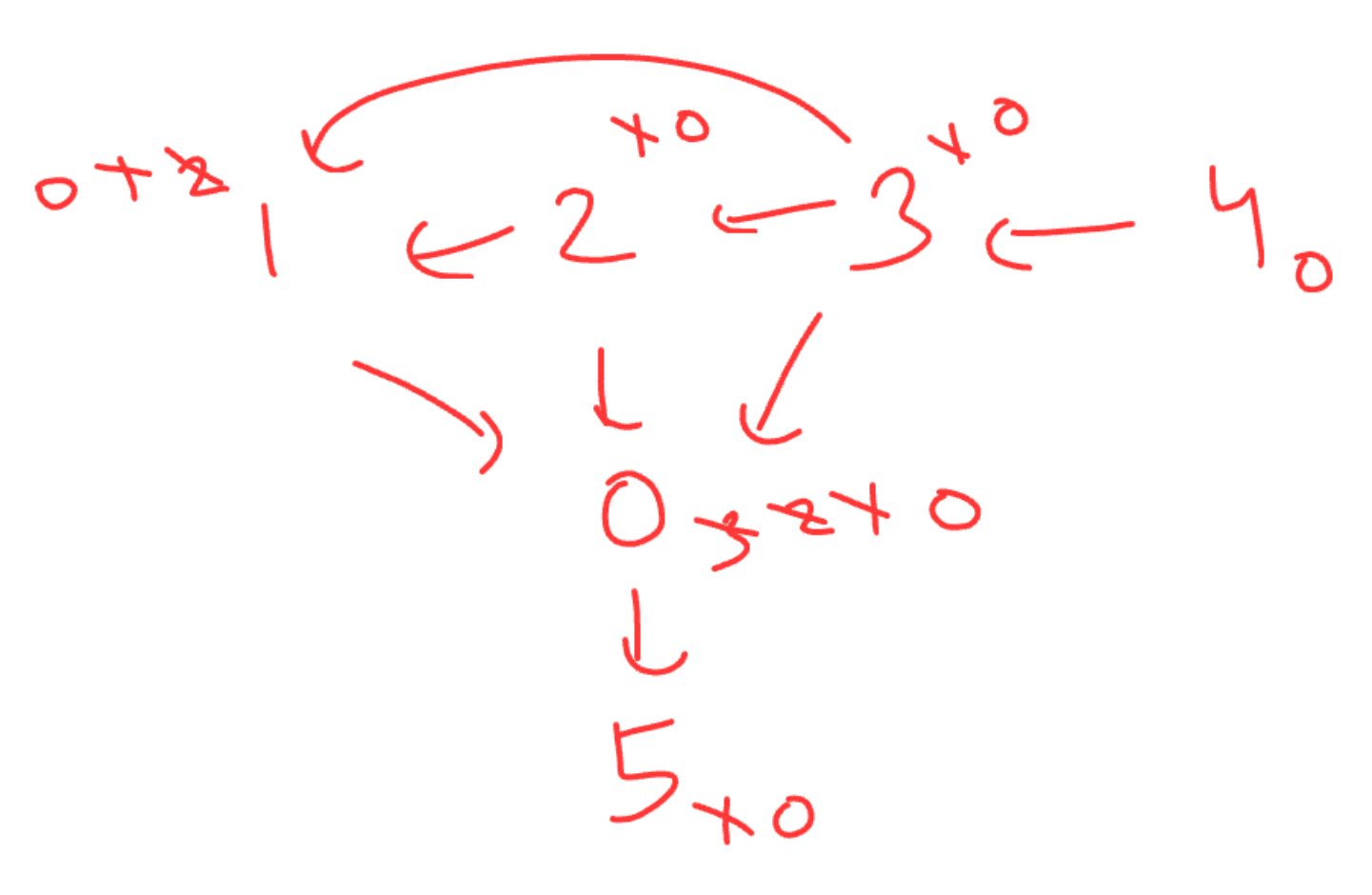



1 (2) ×0
2 (1) 0
3 (1) 0
4 (0)
0 (3) 2 1 0
5 (1) 0

Indegree: No. of incoming edges to a node.

Outdegree: No. of outgoing edges from a node.

4, 3, 2, 1, 0, 5


0 1 2 1
2 1 0
3 1 0
4 0
0 3 2 1 0
5 1 0

TC: O(V + E).

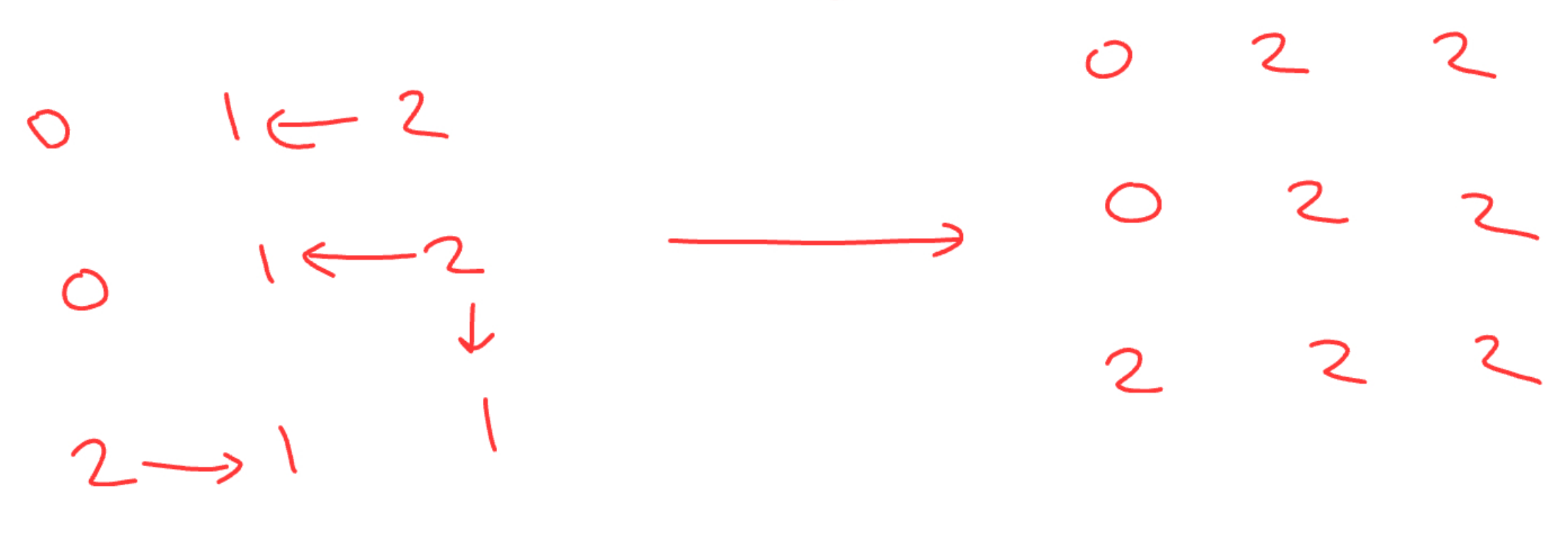
AS: O(V).

2: [0, 1].

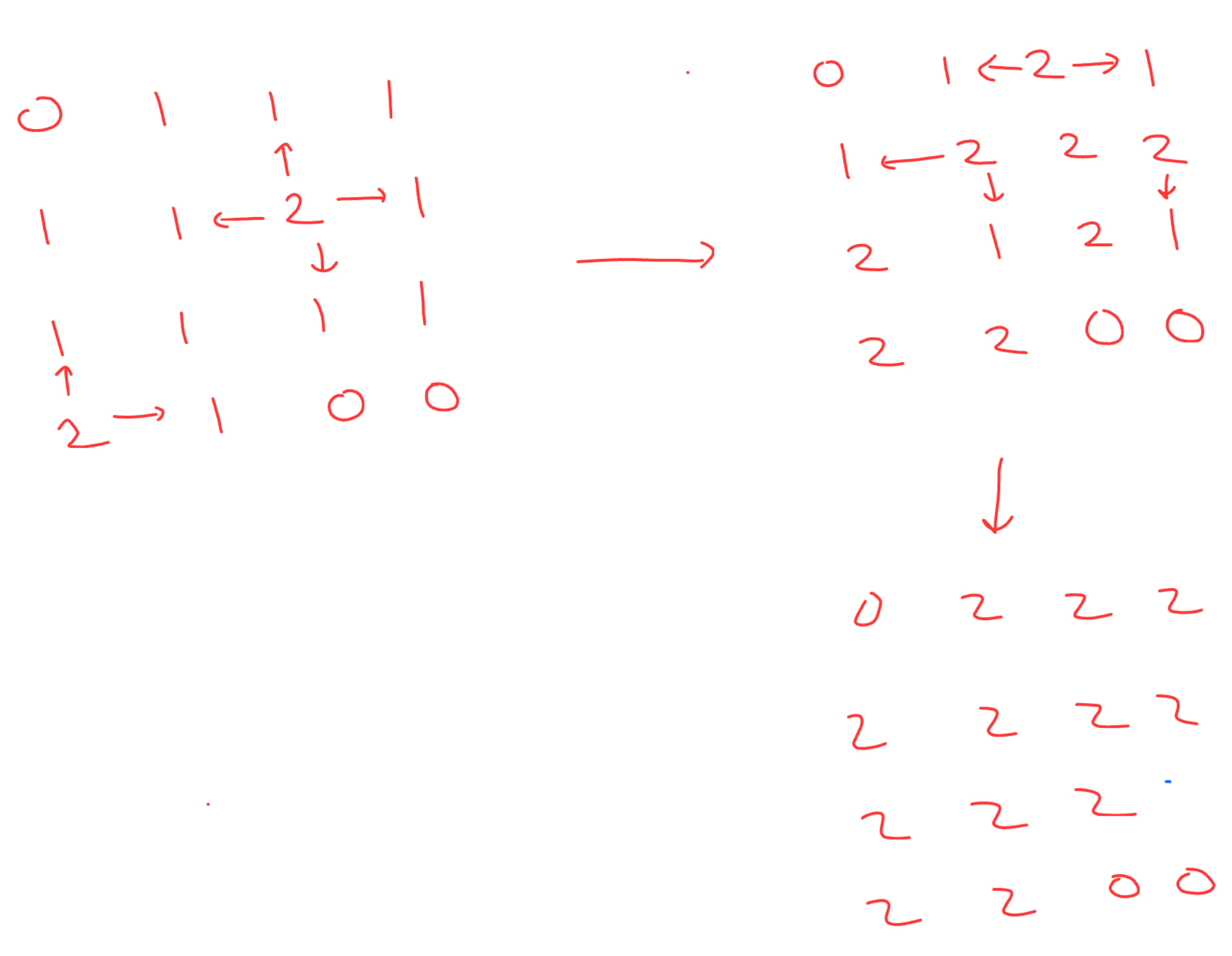
q: ~~4~~ ~~3~~ ~~2~~ ~~1~~ ~~0~~ ~~5~~

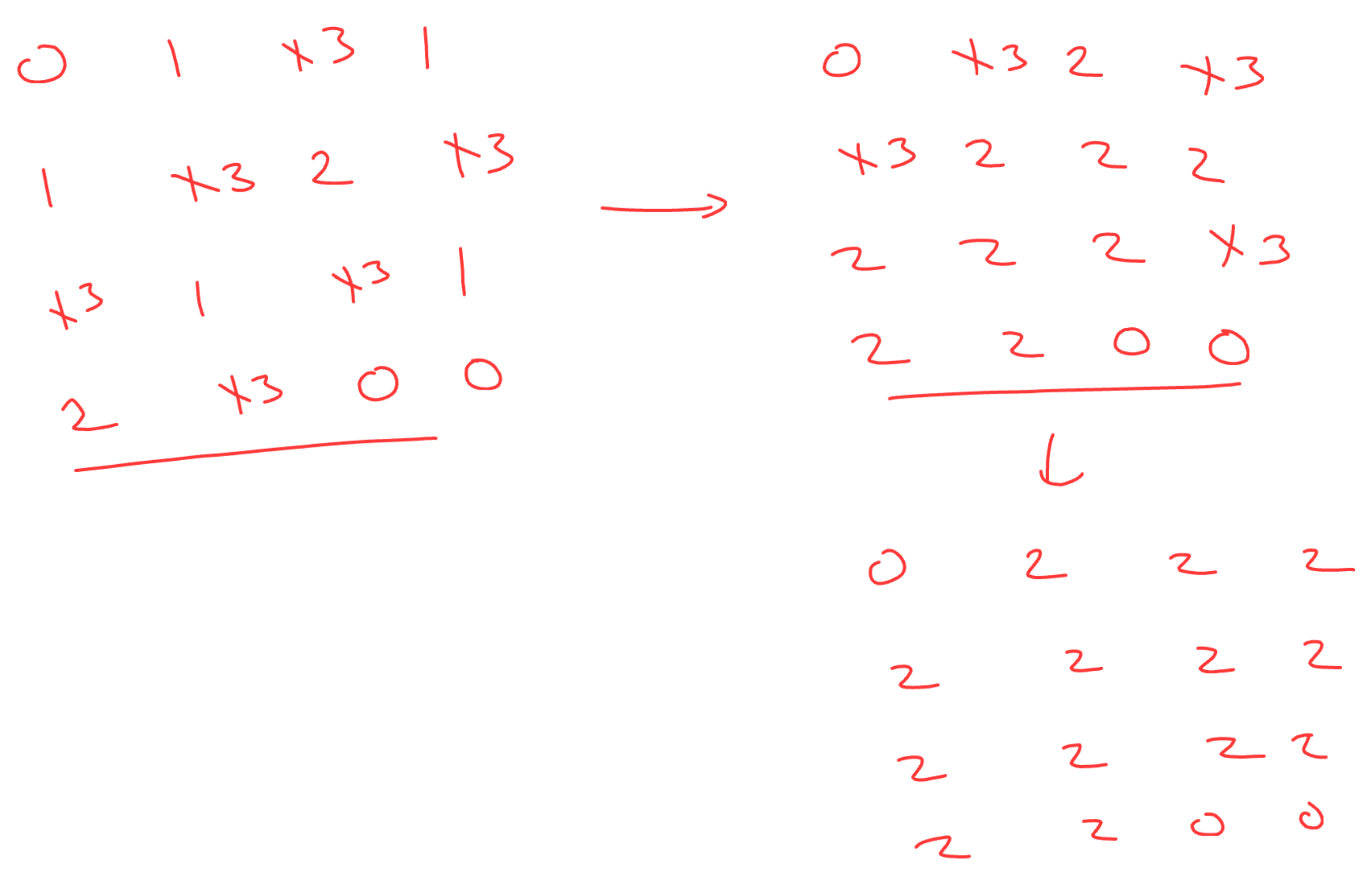
res: 4, 3, 2, 1, 0, 5

Rotten Oranges.


0 1 2
0 1 2
2 1 1
0 2 2
0 2 2
2 2 2

$$\begin{array}{cccc} 0 & 1 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 1 \\ 2 & 1 & 0 & 0 \end{array}$$



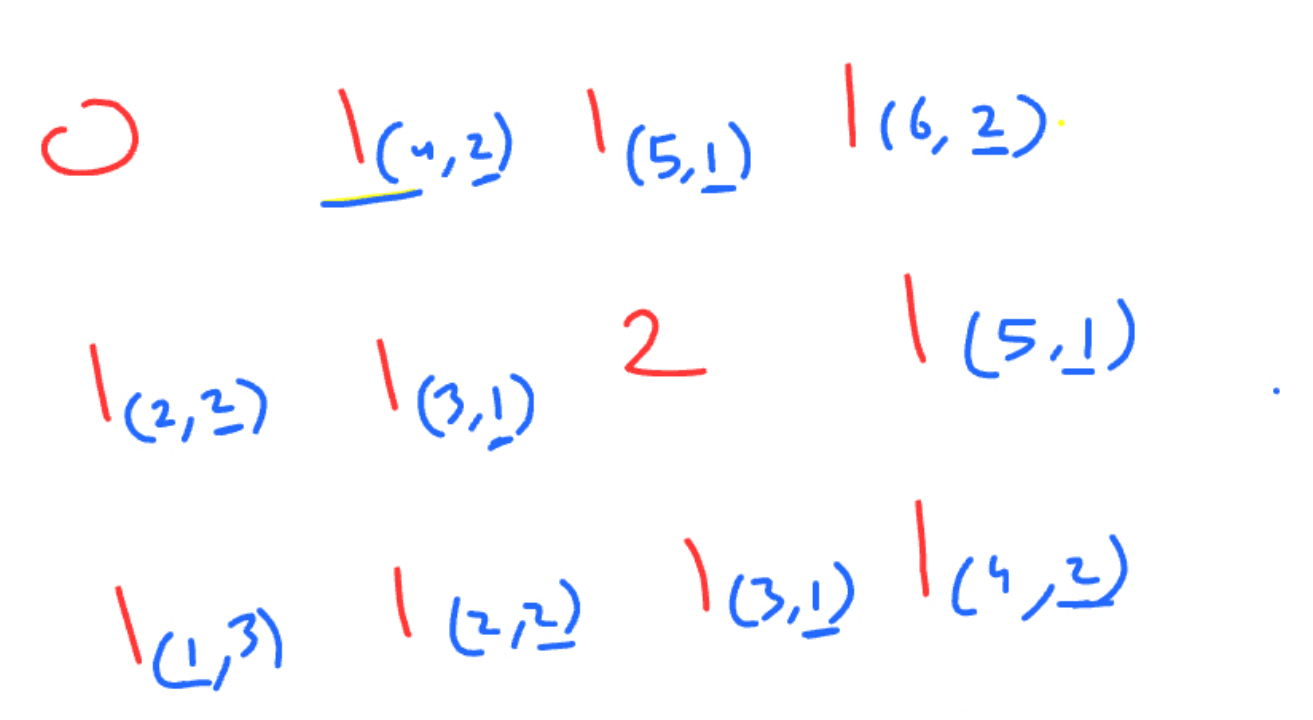


$$\begin{array}{ccccc} 2 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ \textcircled{1} & 1 & 1 & 1 & 1 \end{array}$$

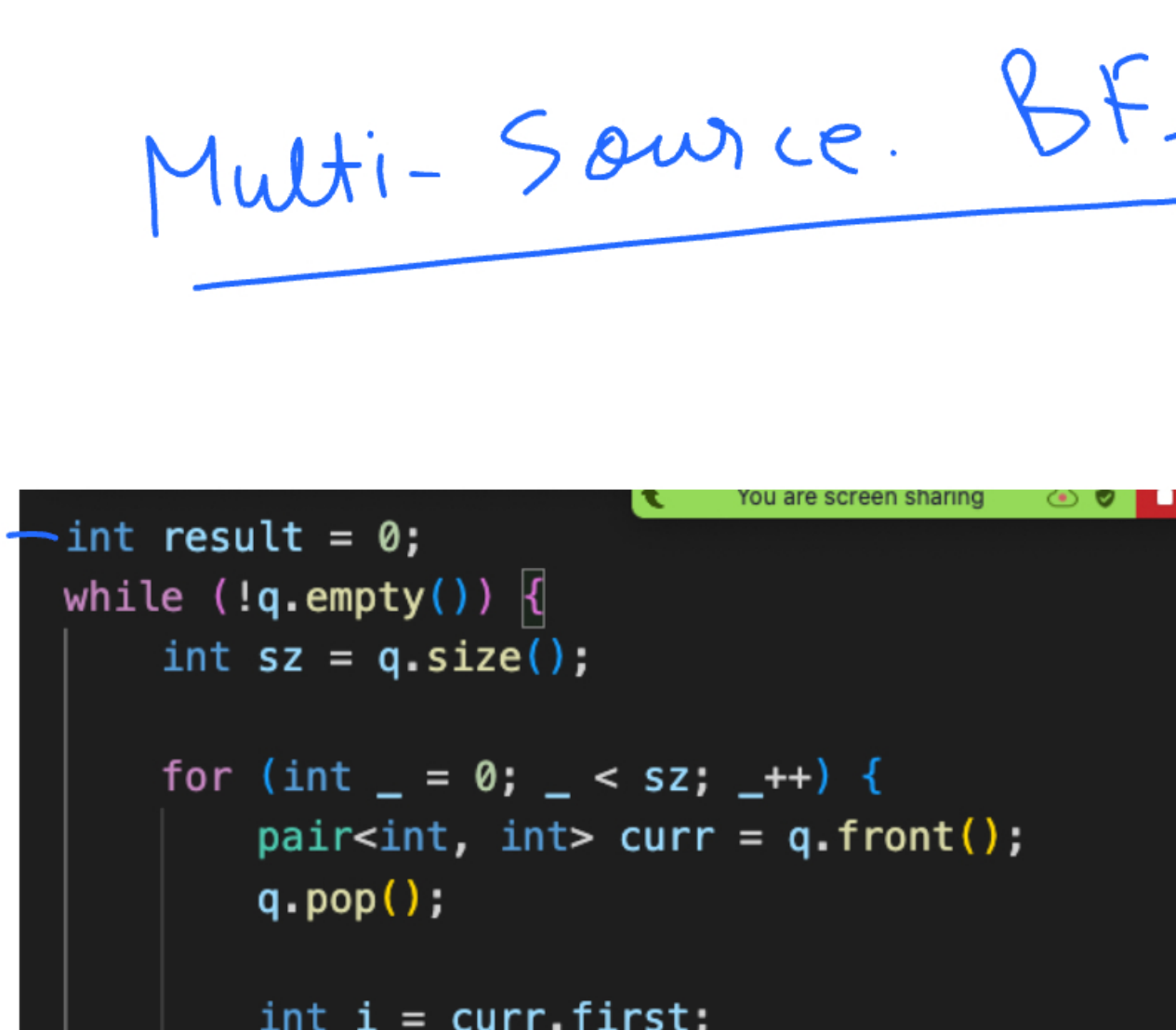
time ~ $O(n^2)$.

BF. TC: $O(n^2 \times n^2) = O(n^4)$.

AS: $O(1)$.


0 1(4,2) 1(5,1) 1(6,2)
1(2,2) 1(3,1) 2 1(5,1)
1(1,3) 1(2,2) 1(3,1) 1(4,2)
2 1(1,3) 0 0

Multi-Source BFS.


You are screen sharing

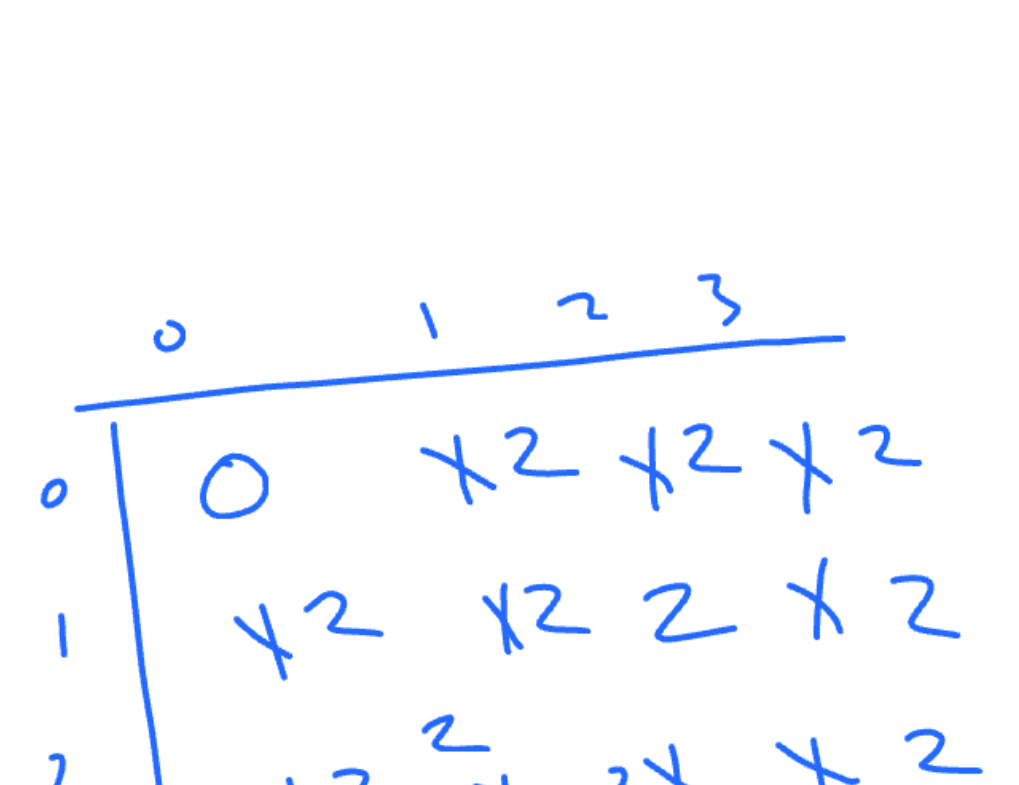
```
int result = 0;
while (!q.empty()) {
    int sz = q.size();

    for (int _ = 0; _ < sz; _++) {
        pair<int, int> curr = q.front();
        q.pop();

        int i = curr.first;
        int j = curr.second;

        for (int k = 0; k < 4; k++) {
            int new_i = i + dx[k];
            int new_j = j + dy[k];

            if (isSafe(new_i, new_j, arr)) {
                arr[new_i][new_j] = 2;
                q.push({new_i, new_j});
            }
        }
    }
    result++;
}
```



res = ~~0~~ ~~1~~ ~~2~~ 3

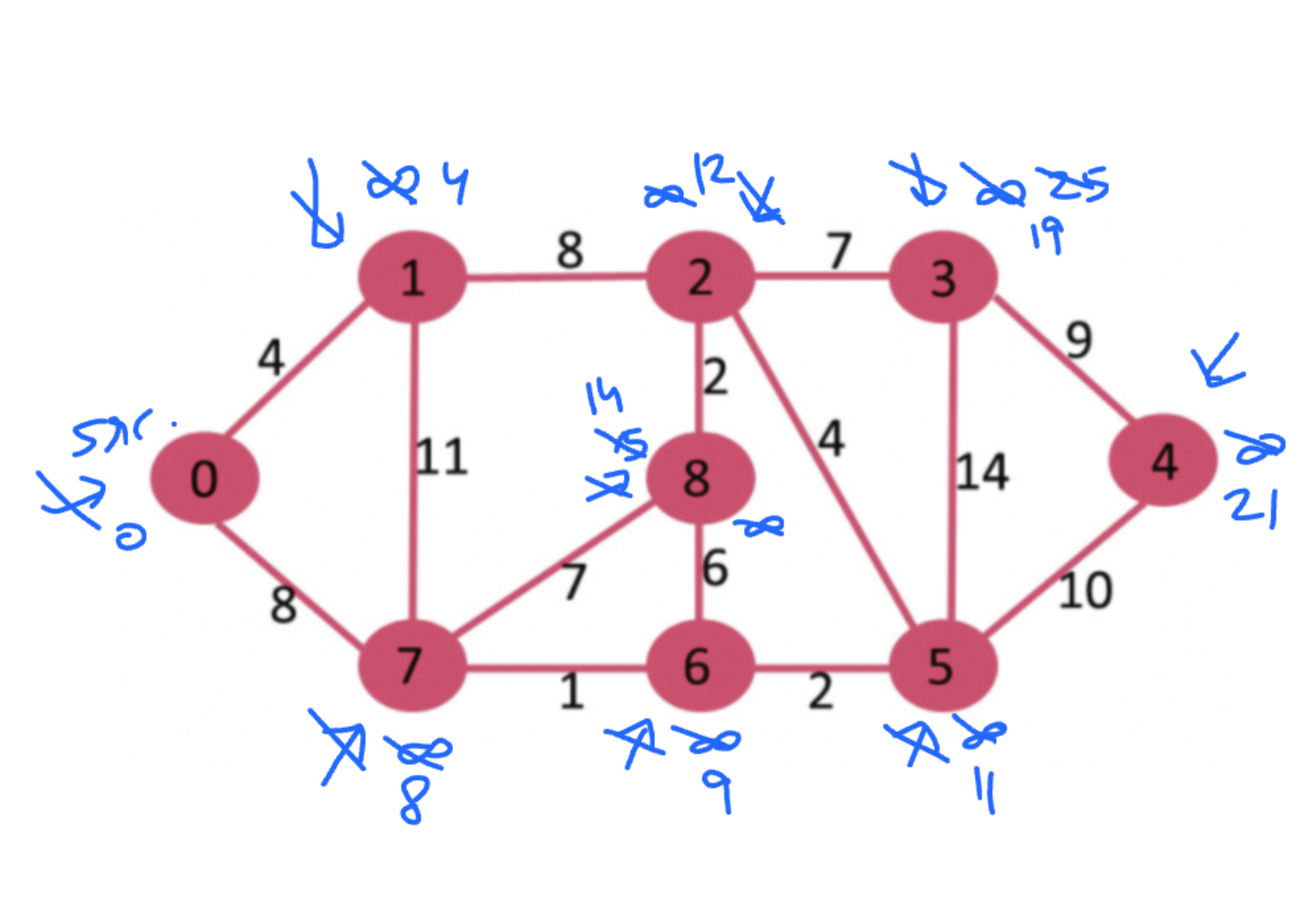
q: ~~(1,2)~~ ~~(3,0)~~ ~~(0,2)~~ ~~(1,1)~~ ~~(1,3)~~ ~~(2,2)~~ ~~(2,0)~~ ~~(3,1)~~ ~~(0,1)~~ ~~(0,3)~~ ~~(1,0)~~ ~~(2,1)~~ ~~(2,3)~~

sz = ~~2~~ ~~6~~ 5

curr = ~~(1,2)~~ ~~(3,0)~~ ~~(0,2)~~ ~~(1,1)~~ ~~(1,3)~~ ~~(2,3)~~ (2,0) ~~(3,1)~~

$$\begin{array}{cc} 2 & 0 \\ 0 & 1 \end{array}$$

Dijikstra


src.
0
1
2
3
4
5
6
7
8
4
8
7
9
14
10
2
1
7
6
2
11
4
0
∞ 4
∞ 12
∞ 25 19
∞ 21
∞ 11
∞ 9
∞ 8
14 15

(distU, U)

↑ ordering on distances

PQ: ~~(0, 0)~~ ~~(4, 1)~~ ~~(8, 7)~~ ~~(12, 2)~~ ~~(15, 8)~~ ~~(9, 6)~~ ~~(11, 5)~~ ~~(25, 3)~~ ~~(21, 4)~~ ~~(14, 8)~~ ~~(19, 3)~~

distU = ~~0~~ ~~4~~ ~~8~~ ~~9~~ ~~11~~ ~~12~~ ~~14~~ ~~19~~ ~~21~~ 25

U = ~~0~~ ~~1~~ ~~7~~ ~~6~~ ~~5~~ ~~2~~ ~~8~~ ~~3~~ ~~4~~ 3